

Then, to compile and install, use the command given below:

```
$ make
$ sudo make install
```

 **Note:** If the target is an embedded one like PPC or ARM, 'make install' is not of any use. Hence you have to identify the necessary executables and library after compilation, and place them in your file system.

Once we have successfully compiled and installed the software, create a *lighttpd* user and group:

```
# groupadd lighttpd
# useradd -g lighttpd -d /home/sganguly/Documents/myserver/
-s /sbin/nologin Lighttpd
```

Here, */home/sganguly/Documents/myserver/* is the root directory for the server.

Next, create a log directory for the server to store logs:

```
# mkdir /var/log/lighttpd
# chown lighttpd:lighttpd /var/log/Lighttpd
```

Now, create a directory for the documents that the HTTP server will be serving (Document root):

```
# mkdir /home/sganguly/Documents/myserver
```

This can be anything of your choice as long as you mention it properly in the configuration file. Now create *Index.html* file with any content, and place it in this directory.

We are now ready to run the Lighttpd server.

```
# /usr/local/sbin/lighttpd -f /home/sganguly/Documents/
lighttpd.conf
```

In the above command, */home/sganguly/Documents/lighttpd.conf* is the configuration file used. The configuration file's content should be as follows:

```
server.document-root = "/home/sganguly/Documents/myserver"

server.port = 80
server.username = "lighttpd"
server.groupname = "lighttpd"
server.bind = "127.0.0.1"
server.tag = "lighttpd"
```

```
server.errorlog = "/var/log/lighttpd/error.log"
accesslog.filename = "/var/log/lighttpd/access.log"
```

```
server.modules = (
    "mod_access",
    "mod_accesslog",
    "mod_fastcgi",
    "mod_rewrite",
    "mod_auth"
)
```

```
# mimetype mapping
mimetype.assign = (
    ".pdf" => "application/pdf",
    ".tar.gz" => "application/x-tgz",
    ".tgz" => "application/x-tgz",
    ".tar" => "application/x-tar",
    ".zip" => "application/zip",
    ".mp3" => "audio/mpeg",
    ".css" => "text/css",
    ".html" => "text/html",
    ".htm" => "text/html",
    ".js" => "text/javascript",
    ".c" => "text/plain",
    ".conf" => "text/plain",
    ".text" => "text/plain",
    ".txt" => "text/plain",
    ".xml" => "text/xml",
    ".mpeg" => "video/mpeg",
)

index-file.names = ( "index.html" )
```

The Lighttpd server will start at 127.0.0.1:80 and, by default, it will load *index.html*. So, go to a browser and type the following command in the browser window:

```
http://127.0.0.1:80
```

Then hit *Enter*, and you will be shown the *index.html* page.

Adding support for fast CGI

To generate dynamic behaviour and server side computation, there are PHP, JSP and many other things. But for an embedded system with limited memory, in order to get fast responses, C is still the best choice. CGI will allow your C code in the server to talk to the HTTP client.

Step 1: Download *fcgi2* from <https://github.com/FastCGI-Archives/fcgi2>.

Step 2: Compile and install.